

PHASE-ANGLE CONTROL OF SCR USING AT89C51

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Silicon-controlled rectifiers (SCR) are solidstate semiconductor devices that are usually used in power switching circuits. SCR controls the output signal by switching it 'on' or 'off', thereby controlling the power to the load in context. The two primary modes of SCR control are phase-angle fired—where a partial waveform is passed every half cycle—and zero-crossing fired—where a portion of

Some of the features of this microcontroller-based phase-angle controller for SCR are:

1. Utilises the zero-crossing detector circuit
2. Controls the phase angle from 0–162°
3. Displays the phase angle on an LCD panel
4. LED indicators are used for displaying the status of SCR
5. Increases or decreases the phase angle with intervals of 18°

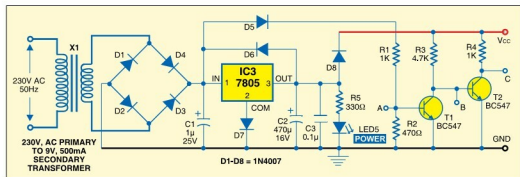


Fig. 1: Power supply and zero-crossing detector circuits

the complete waveforms is passed to regulate the power.

In the phase-angle controller, the firing pulse is delayed to turn on the SCR in the middle of every half cycle. This means that every time a part of an AC cycle is cut, the power to the load also gets cut. To deliver more or less power to the load, the phase angle is increased or decreased, thereby controlling the throughput power.

There are several ways to control the firing angle of SCR. This article describes a microcontroller AT89C51-based phase-angle controller. A microcontroller can be programmed to fire SCR over the full range of half cycles—from 0 to 180°—to get a good linear relationship between the phase angle and the delivered output power.

Basically, the zero-crossing detector circuit interrupts the microcontroller after every 10 ms. This interrupt commands the microcontroller to generate some delay (in the range of 1 ms to 9 ms). The user can increase or decrease the delay in intervals of 1 ms using switches. The SCR is then fired through the opto-coupler. This repeats after every 10 ms.

Circuit description

The complete circuit is divided into two sections:

1. The zero-cross detector section
2. The control section

The zero-cross detector section.

Fig. 1 shows the circuit diagram of the zero-crossing detector and the power supply. The main sections of the circuit

are a rectifier, regulated power supply and zero-crossing detector. The 230V AC mains is stepped down by transformer X1 to deliver the secondary output of 9V, 500 mA. The transformer output is rectified by a full-wave bridge rectifier comprising diodes D1 through D4 and then regulated by IC 7805 (IC3). Capacitors C2 and C3 are used for bypassing the ripples

present in the regulated 5V power supply. A capacitor above 10µF is connected across the output of the regulator IC, while diode D6 protects the regulator IC in case their input is short to ground. LED5 acts as the power-on indicator and resistor R5 limits the current through LED5.

This regulated 5V is also used as biasing voltage for both transistors (T1 and T2) and the control section. A pulsating DC voltage is applied to the base of transistor T1 through diode D5 and resistors R1 and R2. When the pulsating voltage goes to zero, the collector of transistor T1 goes high. This is used for detecting the pulse when the voltage is zero. Finally, the detected pulse from 'C' is fed to the microcontroller of the control section.

The control section. Fig. 2 shows the circuit diagram of the control section for the phase-angle control of SCR. It comprises a microcontroller AT89C51, opto-coupler MCT2E, LCD module and a few discrete components. Port 0 (P0.0 through P0.7) of AT89C51 is used for interfacing data input pins D0 through D7 of the LCD

